DiD: the Good, the Bad, and the Ugly

Lecture 2: Difference-in-Differences

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Today's roadmap

Roadmap

The 2×2 DiD

Controls in DiD: Handle with Care

TWFE with Staggered Adoption: What Goes Wrong

Modern DiD Estimators

Sensitivity Analysis: HonestDiD

Lab Preview

John Snow and the Broad Street pump (1854)

Setting

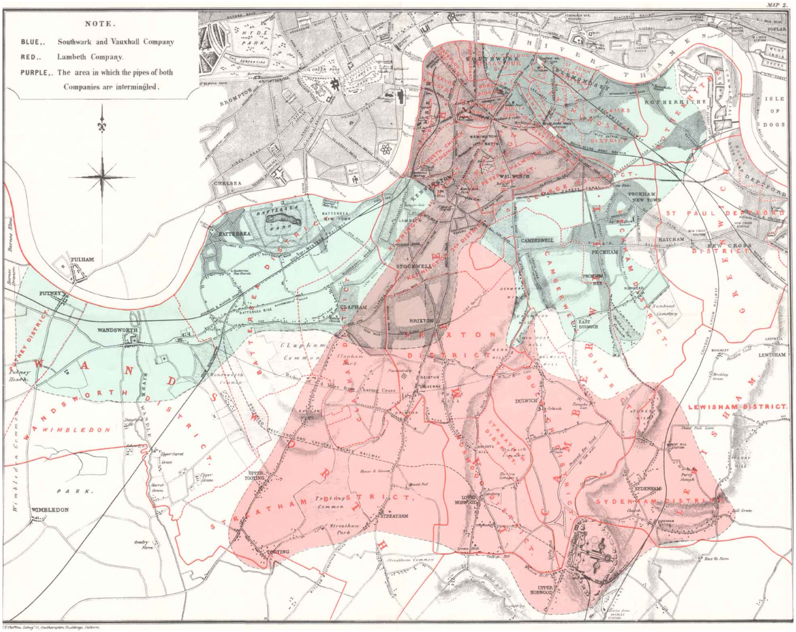
- London cholera epidemic, 1854
- Lambeth Co. moved its water intake upstream (cleaner water) in 1852
- Southwark & Vauxhall Co. kept intake near sewage outflow
- Both companies served *interleaved* streets ⇒ no sorting by class

Why it works

- Common shocks (weather, population density) affect both companies equally
- Only Lambeth changed its water source
- The comparison group approximates the counterfactual trend for Lambeth customers

The DiD table (Deaths per 10 000 houses)

	1849	1854
Southwark & Vauxhall	135	147
Lambeth	130	17
DiD		**−125**


MAP 2.
NOTE.
BLUE,. Southwark and Vauxhall Company
RED.. Lambeth Company.
PURPLE,. The area in which the pipes of both Companies are intermingled.
HYDE PARK
BROMPTON
CHELSEA
BATTERSEA
FULHAM
PUTNEY
WANDSWORTH
WIMBLEDON
CLAPHAM
STOCKWELL
BRIXTON
KENNINGTON
WALWORTH
CAMBERWELL
PECKHAM
ROTHERHITHE
SOUTHWARK
HATCHAM
NEW CROSS
DULWICH
STREATHAM
TOOTING
UPPER TOOTING
SYDENHAM
UPPER NORWOOD
LOWER NORWOOD
LEWISHAM
LEWISHAM DISTRICT.
ISLE OF DOGS
POPLAR
RIVER THAMES
WANDSWORTH DISTRICT.
STREATHAM DISTRICT.
SYDENHAM DISTRICT.
BRIXTON DISTRICT.
WEST END AND CRYSTAL PALACE RAILWAY

Cholera and DiD: compared to what?

1. Different companies

1854 only — naive cross-section

Company	Outcome
Lambeth	$Y = \alpha_L + \delta$
S&V	$Y = \alpha_{SV}$

Estimate: $\hat{\delta} + (\alpha_L - \alpha_{SV})$

Selection bias

The unit fixed effects $\alpha_L - \alpha_{SV}$ are not removed. Lambeth and S&V customers may differ in ways that affect cholera risk independently of the water supply.

2. Before and after

Lambeth only — interrupted time series

Company	Time	Outcome
Lambeth	Before	$Y = \alpha_L$
Lambeth	After	$Y = \alpha_L + \tau + \delta$

Estimate: $\hat{\delta} + \tau$

Time-trend bias

The fixed effect α_L cancels, but the common time shock τ does not. Cholera deaths were rising for S&V too — we cannot distinguish δ from τ.

3. Difference-in-Differences

Both dimensions — the fix

Company	Time	Outcome	Δ_1
Lambeth	Before	α_L	
Lambeth	After	$\alpha_L + \tau + \delta$	$\tau + \delta$
S&V	Before	α_{SV}	
S&V	After	$\alpha_{SV} + \tau$	τ
$\Delta_2 = (\tau + \delta) - \tau$			$\boldsymbol{\delta}$

Why it works

Δ_1 (within-unit) eliminates α. Δ_2 (across units) eliminates τ. What remains is δ — **if** the time shock τ is the same for both units (*parallel trends*).

Card & Krueger (1994): minimum wage and employment

Setting

- New Jersey raised minimum wage from \$4.25 to \$5.05 in April 1992
- Pennsylvania left minimum wage unchanged at \$4.25
- Survey of fast-food restaurants: Feb 1992 (pre) and Nov 1992 (post)

The DiD table

	Pre	Post	Δ
NJ (treated)	20.44	21.03	$+0.59$
PA (control)	23.33	21.17	-2.16
DiD			$+$**2.75**

Full-time equivalent employees per restaurant

The surprise

Standard theory predicted a minimum wage *increase* would *reduce* employment. The DiD estimate is positive and statistically significant.

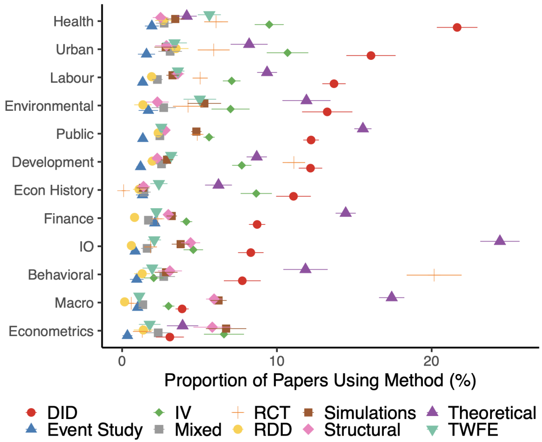
Key identification assumption

NJ and PA fast-food restaurants would have followed the same employment trend absent the NJ wage increase. Their pre-period trends were indeed parallel — but verifying this requires data.

Card and Krueger (1994)

DiD is now the dominant quasi-experimental design

Figure 5: Cross-Sectional Breakdown of Empirical Methods by Field in NBER and CEPR Working Papers



Garg and Fetzer (2025)

The 2×2 setup: four averages and three subtractions

Two groups (k = treated, U = untreated), two periods (Pre, Post).

	Pre	Post	Δ
Treated (k)	$\bar{Y}_k^{pre}$	$\bar{Y}_k^{post}$	$\bar{Y}_k^{post} - \bar{Y}_k^{pre}$
Untreated (U)	$\bar{Y}_U^{pre}$	$\bar{Y}_U^{post}$	$\bar{Y}_U^{post} - \bar{Y}_U^{pre}$
DD			$\hat{\delta}^{DD} = (\bar{Y}_k^{post} - \bar{Y}_k^{pre}) - (\bar{Y}_U^{post} - \bar{Y}_U^{pre})$

OLS representation:

$$Y_{ist} = \alpha + \gamma \cdot \text{Treat}_s + \lambda \cdot \text{Post}_t + \delta \cdot (\text{Treat}_s \times \text{Post}_t) + \varepsilon_{ist}$$

- $\hat{\delta}$ is the coefficient on the interaction term
- Absorbs pre-existing level differences (γ) and common time trends (λ)

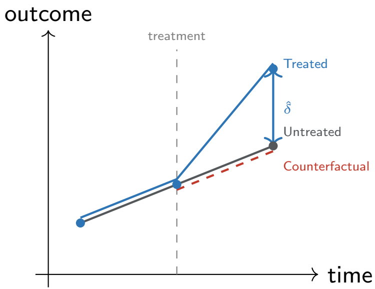
Formal statement

$$E[Y_t^0 - Y_{t-1}^0 \mid D = 1] = E[Y_t^0 - Y_{t-1}^0 \mid D = 0]$$

In the absence of treatment, treated and untreated units would have followed the *same trend*.

What it is not

- Not "same levels" — DiD removes level differences
- Not testable in the post-period (we observe Y^1, not Y^0, for treated)
- Pre-trend tests are informative but not sufficient



Two auxiliary assumptions

No anticipation

Treatment has no effect on potential outcomes before the treatment date.

$$Y_{it}^{1} = Y_{it}^{0} \quad \forall\, t < T^*$$

Violation: firms start adjusting hiring *before* a policy is announced. Pre-treatment event-study coefficients should be zero under this assumption.

SUTVA (Stable Unit Treatment Value Assumption)

- **No interference:** treatment of unit i does not affect outcomes of unit j
- **No hidden versions:** there is a single version of treatment

Violation: a local job-training programme creates spillovers to untreated workers in the same city.

Falsification strategies: making the case for parallel trends

Three complementary tools (Cunningham, 2021):

1. **Event-study pre-trends**: regress on leads and lags of treatment. Pre-treatment coefficients should be statistically indistinguishable from zero.
2. **Same outcome, different group**: apply the same DiD to a group that should be unaffected. If that group also shows an effect, something is wrong. *Example:* Medicaid expansion on the near-elderly vs. the elderly (already on Medicare).
3. **Same group, placebo outcome**: apply the DiD to an outcome that the treatment *cannot* affect. *Example:* test whether a minimum wage affects *pre-reform* employment.

Important

None of these tests *proves* parallel trends — they make it more or less *plausible*. Parallel trends is a fundamentally untestable assumption about an unobserved counterfactual.

From 2×2 to panel: TWFE

With multiple units and periods, estimate:

$$Y_{it} = \alpha_i + \gamma_t + \delta\, D_{it} + \varepsilon_{it}$$

- α_i: unit fixed effects — absorb all time-invariant heterogeneity
- γ_t: time fixed effects — absorb common shocks
- D_{it}: treatment indicator (switches on at treatment date)

Why does this work? Frisch-Waugh-Lovell (FWL)

Absorbing $\alpha_i + \gamma_t$ by demeaning is equivalent to projecting out unit and time means. $\hat{\delta}$ is identified from *within-unit, within-time-period* variation in treatment after removing unit trends and time shocks. This is the **two-way within-group estimator**.

When TWFE = ATT

Under homogeneous and constant treatment effects and a single treatment cohort, TWFE recovers the ATT. These conditions break down with *staggered adoption*.

Roadmap

The 2×2 DiD

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Controls in DiD are not like controls in OLS

In cross-sectional OLS: controls absorb confounders that bias the coefficient of interest.

In DiD: the design already removes time-invariant differences. Controls serve a *different* purpose: they help justify **conditional parallel trends** when unconditional trends differ.

Problem 1 — Bad controls (Huntington-Klein, 2023)

If covariates are **time-varying**, including them in TWFE introduces new bias:

- A post-treatment covariate affected by treatment is a *collider* — conditioning on it opens a backdoor path
- Pre-treatment levels are safer; post-treatment realisations of the same variable are not

Problem 2 — Weighting: from conditional ATT to $\hat{\beta}_{treat}$ (Baker et al., 2026)

Even with correctly chosen pre-treatment covariates (correct, well measured and correct functional form!), $\hat{\beta}_{treat}$ from regression DiD **does not recover** $ATT(2)$ unless treatment effects are constant across covariate strata (additional assumption).

- $\hat{\beta}_{treat}$ is a weighted average of stratum-specific $ATT_{X_i}(2)$ with *non-convex* weights
- Overweights strata rare among treated; underweights strata common among treated
- Can be negative even when every $ATT_{X_i}(2) > 0$

Safer strategies

1. unconditional parallel trends, but is it credible?
2. pre-treatment values only: averages? not too close to the treatement date?
3. propensity-score weighting;
4. doubly-robust CS estimator: targets $ATT(2)$ directly and handles both problems by design but by default it takes the initial value and treats the variable as time invariant

Roadmap

Staggered adoption: the real-world setting

In practice, units often adopt treatment at *different times*:

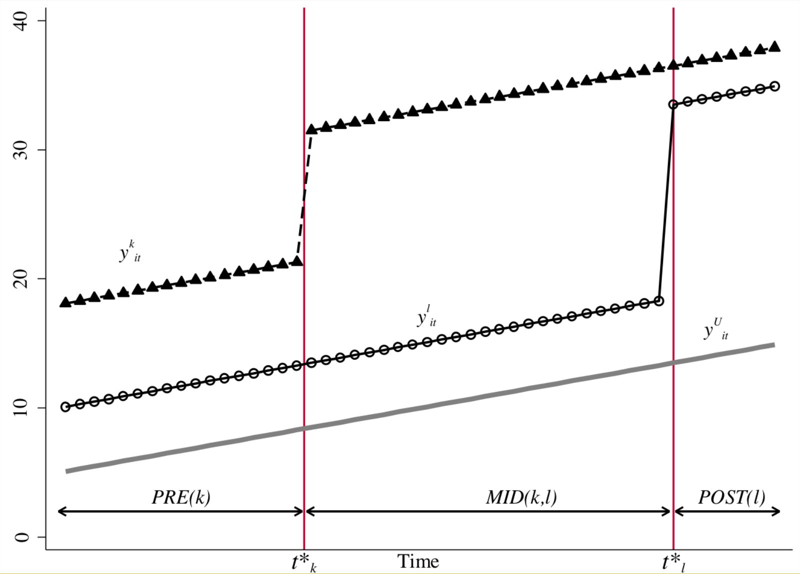
- US states adopt right-to-carry laws in different years
- Firms undergo M&A in different waves
- Firms introduce robots in production processes in different years

With staggered timing and multiple cohorts, TWFE still runs fine — but what does $\hat{\delta}$ identify?

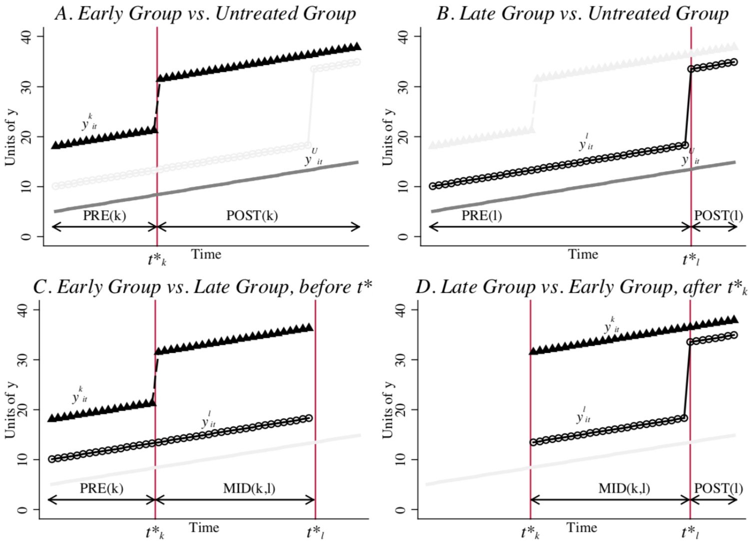
Goodman-Bacon (2021) insight

TWFE with staggered adoption is a **variance-weighted average of all possible** 2×2 **DiDs** — including comparisons that use *already-treated* units as controls. These “forbidden comparisons” can produce negative weights and contaminate the estimate.

TWFE comparisons



TWFE comparisons



The Bacon decomposition

With treatment groups k (early) and l (late) plus never-treated U, TWFE decomposes as:

$$\hat{\delta}^{TWFE} = \sum_{k \neq U} s_{kU}\, \hat{\delta}_{kU}^{2\times 2} + \sum_{k \neq U} \sum_{l > k} s_{kl} \left[\mu_{kl}\, \hat{\delta}_{kl}^{2\times 2,k} + (1 - \mu_{kl})\, \hat{\delta}_{lk}^{2\times 2,l} \right]$$

- $\hat{\delta}_{kU}^{2\times 2}$: clean — early vs. never-treated (safe comparison)
- $\hat{\delta}_{kl}^{2\times 2,k}$: clean — early vs. not-yet-treated late group (safe)
- $\hat{\delta}_{lk}^{2\times 2,l}$: dangerous — late group vs. *already-treated* early group

The dangerous term picks up:

$$\hat{\delta}_{lk}^{2\times 2,l} = ATT_{l,Post(l)} + \underbrace{\Delta Y_l^0 - \Delta Y_k^0}_{\text{PT bias}} - \underbrace{(ATT_k^{Post} - ATT_k^{Mid})}_{\text{Heterogeneity bias}}$$

If treatment effects *grow over time* ($ATT_k^{Post} > ATT_k^{Mid}$), the heterogeneity bias is negative, and TWFE can flip sign.

A complementary lens: negative implicit weights

Rewrite TWFE not as 2×2 DDs but as a weighted sum of *cell-level* treatment effects (de Chaisemartin and D'Haultfœuille, 2020):

$$\hat{\delta}^{\,TWFE} = \sum_{\ell,t} w_{\ell t} \cdot ATT(\ell, t)$$

The problem

Some $w_{\ell t}$ are **negative** → TWFE *subtracts* positive treatment effects from the estimate.

Why negative?

- FE demeaning uses already-treated units as a "control" for later cohorts
- Their demeaned treatment indicator is *below* the unit mean in later periods → enters the regression with a negative sign
- Result: large late-period ATTs of early-treated cohorts get subtracted from $\hat{\delta}^{\,TWFE}$

Note: same pathology as Bacon's heterogeneity bias — two descriptions of "already-treated as control".

Baker (2022) simulation: sign flip from TWFE

Design: 1000 firms, 4 staggered cohorts, 30 years. True average ATT $= +82$. Dynamic treatment effects (effects grow over time).

Estimator	$\hat{ATT}$	Status
True (weighted avg.)	$+82$	benchmark
TWFE	-6.69^{***}	sign flipped
Callaway & Sant'Anna	$+68.3^{***}$	close to truth
Sun & Abraham	$+68.3^{***}$	close to truth

Bacon decomposition of $\hat{\delta}^{TWFE} = -6.69$:

Comparison	Weight	Avg. DD
Early vs. late (clean)	0.50	$+51.8$
Late vs. early (dirty)	0.50	-65.2
Weighted sum		-6.69

Key lesson

The sign flip is not a data problem — it is an *estimator* problem. TWFE assigns 50% weight to a contaminated comparison. Modern estimators fix this by construction.

Roadmap

The forward-engineering response

The Bacon decomposition *diagnoses* the problem but does not solve it. A new generation of estimators avoids the forbidden comparisons by design.

Callaway & Sant'Anna (2021)	**Sun & Abraham (2021)**	**de Chaisemartin & D'Haultfœuille (2020)**
Group-time ATT(g,t); aggregates via user-chosen weights	Interaction-weighted estimator; cohort-specific coefficients	"Switchers" approach; works with multi-valued and reversible treatment
Never- or not-yet-treated comparison; doubly robust	Last-cohort or never-treated comparison	Comparison across switchers and non-switchers

Callaway & Sant'Anna (2021): identification and IPW estimator

Target: $ATT(g,t) = E[Y_t^1 - Y_t^0 \mid G_g = 1]$, where $G_g = 1 =$ units first treated at g.

Assumptions: absorbing treatment · conditional parallel trends (given X) · no anticipation · common support

IPW estimator — reweight clean controls to resemble cohort g:

$$\widehat{ATT}(g,t) = \underbrace{E\left[\frac{G_g}{E[G_g]}\,(Y_t - Y_{g-1})\right]}_{\text{(A) cohort } g \text{ mean change}} - \underbrace{E\left[\frac{w_g(X)}{E[w_g(X)]}\,(Y_t - Y_{g-1})\right]}_{\text{(B) reweighted controls mean change}}$$

$$w_g(X) = \frac{\hat{p}_g(X)\,(1-D_t)(1-G_g)}{1-\hat{p}_g(X)}, \qquad \hat{p}_g(X) = \widehat{P}(G_g = 1 \mid X)$$

- $(1-D_t)(1-G_g)$: selects *clean* controls only — untreated at t, not in cohort g
- $\hat{p}_g/(1-\hat{p}_g)$: odds-ratio weight — upweights controls whose X profile matches cohort g
- **Dividing by** $E[w_g]$: normalises weights to sum to 1, so (B) is a proper average

Callaway & Sant'Anna (2021): aggregation choices

All aggregations are weighted averages of $ATT(g,t) \rightarrow$ different choices weight cohort $\times$ period cells differently.

- **Simple** (uniform): one overall ATT
- **Event-time**: ATT at each relative period $\ell = t - g \rightarrow$ the event-study plot
- **Calendar**: ATT by calendar year t
- **Group**: ATT by cohort $g \rightarrow$ reveals heterogeneity across adoption waves

Contrast with TWFE

TWFE implicitly weights all $ATT(g,t)$ cells, often with negative weights for early adopters. CS lets you choose weights *explicitly* and test whether effects vary across groups.

Key idea: Augment TWFE by interacting treatment dummies with cohort indicators. This saturates the model and recovers cohort-specific event-study coefficients.

Model:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{g} \sum_{\ell \neq -1} \delta_{g,\ell} \left(\mathbf{1}[G_i = g] \times D_{it}^{\ell} \right) + \varepsilon_{it}$$

where $D_{it}^{\ell} = \mathbf{1}[t - G_i = \ell]$ is the relative-time indicator.

Aggregation: average $\hat{\delta}_{g,\ell}$ across cohorts, weighting by cohort size. The resulting event-study plot is free of contamination from other cohorts' treatment effects.

Comparison with CS

SA and CS produce numerically very similar event-study plots when using the same comparison group. SA is more natural when you want the TWFE regression as the starting point; CS is more flexible (doubly robust, can use not-yet-treated, forces time invariant controls).

de Chaisemartin & D'Haultfœuille (2020): brief overview

Setting: treatment can be multi-valued *and* reversible — more general than CS/SA.

Key idea: focus on "switchers" — units whose treatment *changes* between consecutive periods. Compare the outcome change of switchers to the outcome change of non-switchers in the same period.

Identifying assumption: in the absence of the treatment change, switchers and non-switchers would have had parallel trends in that period pair.

When to use dCdH

- Treatment can turn off (job loss, de-regulation)
- Continuous or ordinal treatment intensity
- You want period-by-period robust estimates

When to use CS/SA instead

- Binary absorbing treatment
- You want cohort-level heterogeneity
- Long panels with many pre-periods

At what level should you cluster?

Cluster at the level at which treatment is assigned — *not* at the observation level.

- Policy varies at state level $\rightarrow$ cluster by state
- M&A event varies by firm $\rightarrow$ cluster by firm
- DoE treatment varies by department $\rightarrow$ cluster by department

The few-clusters problem

Standard cluster-robust SEs require many clusters (rule of thumb: $\geq$ 30–50). With few treated clusters, SEs are downward biased and rejection rates inflated.

Solutions: wild cluster bootstrap; randomisation inference; or report both cluster-robust and wild-bootstrap CIs.

The CS and SA estimators use the *influence function* approach for SEs — robust to heteroskedasticity and correctly accounts for estimation of the propensity score.

Roadmap

The pre-trends test is not enough

Event-study pre-treatment coefficients being "flat" is reassuring but insufficient (Roth, 2022).

Three fundamental limitations:

1. **Parallel pre-trends $\neq$ parallel counterfactual post-trends.** A simultaneous policy change in the post-period could create parallel pre-trends but non-parallel post-trends.
2. **Low power.** Pre-trend tests are typically underpowered. We can fail to detect economically meaningful pre-trends (fail to detect type II error, false negative). Roth (2022): in papers from *AER*, *AEJ Applied*, *AEJ Policy*, the pre-trend test had only 50–80% power against violations that would substantially bias the estimate.
3. **Pre-testing distortions.** Conditioning on a non-significant pre-trend introduces a form of *selection bias*: we are analysing a selected subsample of draws where the noise happened to look flat, which inflates the post-treatment estimate.

Rambachan & Roth (2023): a more credible approach

Core idea: Rather than testing whether trends are exactly parallel, impose *restrictions on how far they can deviate*.

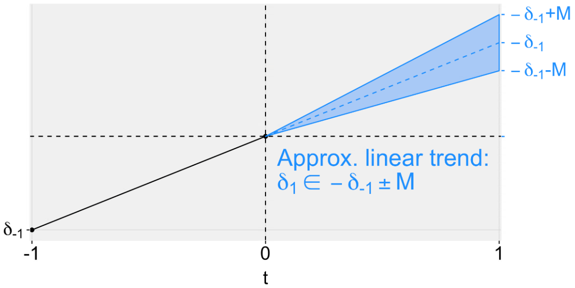
Let δ_t denote the violation of parallel trends in period t:

$$\delta_t = E[Y_t^0 - Y_{t-1}^0 \mid D = 1] - E[Y_t^0 - Y_{t-1}^0 \mid D = 0]$$

We observe $\hat{\delta}_{pre}$ (pre-treatment trend). We do not observe δ_{post}.

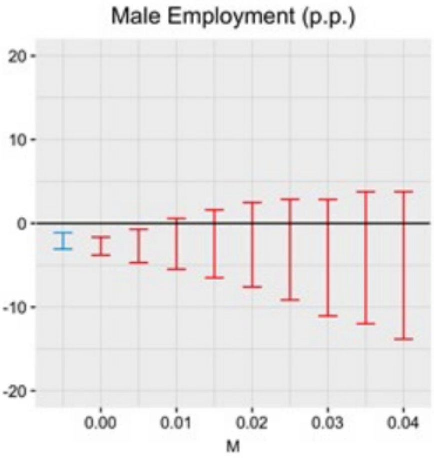
Restriction options:

- **Smoothness (M):** the post-trend cannot deviate more than M from a linear extrapolation of the pre-trend
$$|\delta_{post} - \text{linear extrapolation of } \hat{\delta}_{pre}| \leq M$$
- **Relative magnitudes ($\bar{M}$):** $|\delta_{post}| \leq \bar{M}\, |\delta_{pre}|$



- **Black dot + line:** observed pre-trend δ_{-1}, running from $t = -1$ to $t = 0$
- **Dashed centre line:** pure linear extrapolation into the post period — if the trend were exactly linear it would reach $-\delta_{-1}$ at $t = 1$
- **Blue cone:** all admissible post-trends under the smoothness restriction

$$\delta_1 \in [-\delta_{-1} - M, \ -\delta_{-1} + M]$$



outcome: male employment, p.p.

- **X-axis:** M — the maximum allowed deviation from a linear pre-trend extrapolation
- **Blue bar ($M \approx 0$):** standard CI under exact parallel trends — narrow, entirely below zero; effect is negative and significant
- **Red bars ($M > 0$):** robust CIs widen as M grows and cross zero already at $M \approx 0.01$

Paper — Robot adoption and product innovation

The question

Does adopting industrial robots enable **product innovation**, or does it crowd it out through diseconomies of scope?

Data

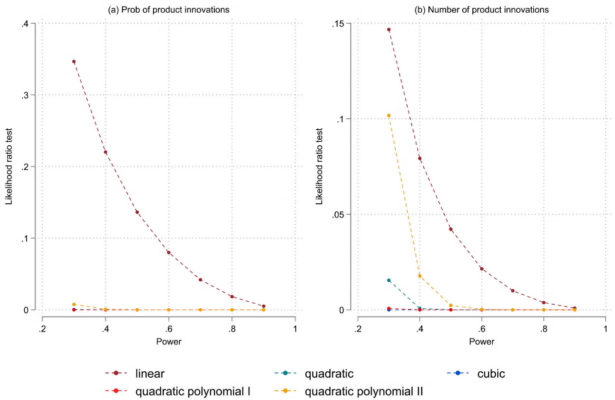
- Spanish firm-level panel (ESEE survey)
- Staggered adoption timing: treated firms adopt robots at different points in time
- Outcomes: probability and count of new product introductions

Key findings

- *Negative* effect of robot adoption on product innovation
- Larger for established, large, non-high-tech firms
- Mechanism: *diseconomies of scope* — robots divert resources toward capacity expansion
- Staggered DiD + IV robustness check

Antonioli et al. (2024) — Antonioli, Marzucchi, Rentocchini & Vannuccini, Research Policy 53 (2024)

Robots paper: pre-trend sensitivity (Roth test)



Reading the plot

- **Y-axis:** LR $= \mathcal{L}(\hat{\beta}_{\text{pre}} \mid \text{violation}) / \mathcal{L}(\hat{\beta}_{\text{pre}} \mid \text{PT})$
  High $\Rightarrow$ data consistent with a violation
- **X-axis:** power to detect that violation

Takeaways

- Non-linear: LR $\approx 0 \rightarrow$ ruled out
- Linear: LR ≈ 0.12 at low power, $\rightarrow 0$ above power 0.5
- Panel (b) drops faster: count outcome safer than probability

Verdict

Only small linear violations survive; parallel trends credible.

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Paper 1 — Academic excellence and faculty recruitment

The question

Did Italy's **Departments of Excellence** (DoE) initiative — a competitive excellence grant awarded in 2018 — raise faculty recruitment in winning departments?

Data

- 290 university departments, balanced panel 2013–2020
- Single treatment cohort: award year 2018
- Outcome: new hires (permanent academic staff)

Rizzo et al. (2025) — Oxford Economic Papers, 2026

Key findings

- Positive average hiring effect
- Driven by *second-tier* departments, not top ones
- More external (inter-institutional) recruitment
- Equalising, not concentrating, effect

Paper 2 — M&As, innovation and superstar firms

The question

Do technology M&As contribute to the rise of **superstar firms** by raising market power, or are they mainly about synergies and innovation?

Data

- 7,978 publicly listed firms, 2008–2020
- M&A deals: Zephyr (Bureau van Dijk)
- Innovation: ORBIS IP patent database
- Outcome: firm-level markup (revenue/variable cost)

Key findings

- +2% average markup increase post-deal
- Mechanism: *patent insulation*, not synergies
- Larger effects for top R&D investors, US firms, high-tech, and AI/GPT deals
- Staggered DiD: Callaway–Sant'Anna & Sun–Abraham

Rentocchini (2025) — Martinez Cillero, Napolitano, Rentocchini, Seri & Zaurino (working paper)

Lab 2: two applied examples

Exercise 1 — Departments of Excellence

Methods:

- TWFE static & event study
- Conditional parallel trends test

Exercise 2 — Superstar M&A

Methods:

- TWFE staggered
- CS & SA combined event-study plot
- Frenchies estimator (dCdH)

Script: `Lab/L2_did.do`

References I

Antonioli, D., Marzucchi, A., Rentocchini, F., and Vannuccini, S. (2024). Robot adoption and product innovation. *Research Policy*, 53(6):105002.

Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., and Sant'Anna, P. H. (2026). Difference-in-differences designs: A practitioner's guide. *Journal of Economic Literature*. Forthcoming.

Card, D. and Krueger, A. B. (1994). Minimum wages and employment: A case study of the fast-food industry in new jersey and pennsylvania. *American Economic Review*, 84(4):772–793.

Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press. Freely available at https://mixtape.scunning.com.

de Chaisemartin, C. and D'Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9):2964–2996.

Garg, P. and Fetzer, T. (2025). Causal claims in economics. *arXiv preprint arXiv:2501.06873*.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2):254–277.

References II

Huntington-Klein, N. (2023). Controls in difference-in-differences don't just work. Data, On Average (Substack). Available at: https://nickchk.substack.com/p/controls-in-difference-in-differences.

Rentocchini, F. (2025). Superstar m&a: Technology acquisitions and firm performance. Working paper.

Rizzo, U., Rentocchini, F., Seeber, M., and Ramaciotti, L. (2025). Beyond the badge of honour: The effect of the academic excellence initiative on staff recruitment in italy. *Oxford Economic Papers*. Forthcoming.

Roth, J. (2022). Pre-test with caution: Event-study estimates after testing for parallel trends. *American Economic Review: Insights*, 4(3):305–322.